SHREYA PITHVA

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Research Interests

Quantum Field Theory, Quantum Simulation, Quantum Computing, Quantum Many-Body Physics, Tensor Networks, Quantum Information Theory and Algorithms, Open Quantum Systems, Quantum Noise and Decoherence and Light–Matter Interaction.

Education

Master of Science in Physics August 2026 – Present
Indian Institute of Science (IISc), Bengaluru
Expected Graduation: May 2027

Bachelor of Science (Research) October 2022 – May 2026
Major: Physics, Minor: Quantum Technology
Indian Institute of Science (IISc), Bengaluru
GPA: 8.33/10

Class 12 (CBSE Central board of Secondary Education, India) Year: 2022
Silver Bells Public School Bhavnagar, Gujarat
Overall percentage: 94.4%

Class 10 (CBSE Central board of Secondary Education, India) Year: 2020
Silver Bells Public School Bhavnagar, Gujarat
Overall percentage: 90.2%

Research Experience

BS Thesis — Tensor Network Methods for (1+1) D Scalar Field Theory
Supervisor: Prof. Aninda Sinha (CHEP, IISc)

- Investigated interacting scalar ϕ^4 field theory in (1+1) dimensions using tensor-network methods, focusing on ground-state properties and quasiparticle excitations.
- Implemented Density Matrix Renormalization Group (DMRG) and infinite DMRG (iDMRG) using the Python-based TeNPy library, employing Matrix Product States (MPS) to study finite and infinite chains of the AKLT model, spin-1 Heisenberg model, and ϕ^4 theory.
- Studied numerical convergence with respect to bond dimension, system size, and local bosonic Hilbert-space truncation.
- Computed physical observables, including ground-state energy density, entanglement entropy, field expectation values $\langle\phi\rangle$ and $\langle\phi^2\rangle$, and correlation lengths.
- Investigated elementary excitations using momentum-dependent variational MPS methods and studied their dispersion relations.
- Currently working on simulating quasiparticles scattering in the spin-1 Heisenberg chain using tensor-network methods, with the aim of extending the approach to ϕ^4 theory.

Term Project — Quantum Simulation of Light–Matter Interaction Using the Jaynes–Cummings and Quantum Rabi Models
Course: Quantum Field Theory on a Quantum Computer
Instructor: Prof. Aninda Sinha (CHEP, IISc)

- Studied the interaction between a quantized electromagnetic field mode and a two-level quantum system using the Jaynes–Cummings and quantum Rabi models.
- Implemented Quantum Phase Estimation (QPE) using QISKIT package in Python to compute energy spectra and compared the results with exact numerical diagonalization.
- Investigated weak, strong, and ultrastrong coupling regimes, focusing on the breakdown of the rotating-wave approximation (RWA), Bloch–Siegert shift, and virtual photons.
- Studied numerical convergence and the effects of photon-number truncation on simulation accuracy.
- Used the Variational Quantum Eigensolver (VQE) for estimating ground-state energies and compared the results with exact calculations.
- Studied light–matter dynamics using second-order Suzuki–Trotter decomposition, including Rabi oscillations, population inversion, and collapse-and-revival phenomena.
- Investigated the effects of depolarizing noise on coherent oscillations and revival phenomena.

Research Project — Multiwavelength Analysis of Supermassive Black Holes

Supervisor: Prof. Prajval Shastri (Raman Research Institute, Bengaluru)

- Conducted computational analysis and a literature review of accreting supermassive black holes and their interactions with the surrounding galactic environment.
- Studied over 100 nearby galaxies at redshifts around $z \approx 0.02$, focusing on active galactic nuclei (AGN) and their host galaxies.
- Analyzed multiwavelength observational data from the S7 survey to investigate AGN activity and synchrotron-emitting jets.
- Explored the influence of AGN jets and accretion processes on the surrounding galactic environment.
- Presented the research as a poster at the Astronomical Society of India (ASI) 2025 conference.
- Link: https://astron-soc.in/asi2025/abstractdetails/ASI2025_89.

Reading and Experimental Project — Nonlinear Photonics and Raman Fibres

Supervisor: Prof. Superdeep (CENSE, IISc)

- Studied fundamental concepts of nonlinear optics, including nonlinear light propagation in optical fibres, Raman scattering, Stokes shifts, and spontaneous parametric down-conversion (SPDC) in nonlinear crystals.
- Gained introductory hands-on experience in optical fibre splicing, laser alignment, and interferometer setup.
- Learned to arrange optical components on an optical table and align lasers for interferometry experiments.
- Gained practical experience with laboratory equipment and basic experimental techniques in optics and photonics.

Reading Project — Modified Newtonian Dynamics (MOND)

Supervisor: Prof. Nirmal Raj (CHEP, IISc)

- Studied the theoretical framework of Modified Newtonian Dynamics (MOND), focusing on modifications to Newtonian gravity in the low-acceleration regime.
- Explored the MOND acceleration scale, interpolation functions, and their implications for gravitational dynamics.
- Investigated the application of MOND to galaxy rotation curves, including the predicted relationship between baryonic mass distribution and rotational velocity.
- Reviewed observational tests of MOND, including the baryonic Tully–Fisher relation and the radial acceleration relation.
- Examined the limitations of MOND in explaining galaxy clusters, gravitational lensing, and large-scale cosmological observations.

On Going Research

Master's Thesis — Quantum Information Scrambling and Hayden–Preskill Recovery

Supervisor: Prof. C. M. Chandrashekhar (DESE, IISc) Sep 2026 – Present

- Investigating quantum information scrambling and the Hayden–Preskill information recovery protocol in quantum many-body systems, with connections to the black-hole information paradox.
- Studying scrambling dynamics and out-of-time-ordered correlators (OTOCs) as diagnostics of quantum chaos, operator growth, and information propagation.
- Exploring theoretical and numerical approaches to understanding the spreading and recovery of quantum information in interacting quantum systems.
- Focused on: Scrambling Dynamics and Out-of-Time-Ordered Correlators in Quantum Many-Body Systems.

S. N. Bhatt Memorial Excellence Fellowship (ICTS) — Quantum Non-Demolition Measurements

Supervisor: Prof. Abhishek Dhar, International Centre for Theoretical Sciences (ICTS–TIFR) May 2026 – Present

- Studying quantum non-demolition (QND) measurements of photon-number distributions in a cavity using atoms as probes.
- Investigating how repeated indirect measurements update photon-number probability distributions through conditional probability updates.
- Exploring the convergence of repeated QND measurements and wave-function collapse, focusing on how successive measurements reveal information about the cavity state.
- Focused on: Convergence of repeated quantum nondemolition measurements and wave-function collapse.

Preprints

Dhruv Shah, R.K. Shishir, Manjaree, Shreya Pithva, TY Booritth Balaji, Rahul Agarwal Singh.
Non-Gaussian statistics in static and dynamics Galton Boards. (also linked to NASA ADS)
https://doi.org/10.48550/arXiv.2404.19478

Relevant Coursework (IISc)

Physics: Quantum Mechanics I & II, Classical Mechanics, Statistical Mechanics, Electromagnetism and the Quantum Physics of Radiation, Thermal Physics and the Physics of Materials, Advanced Mathematical Physics, Quantum Field Theory on a Quantum Computer, Nonlinear Dynamics of Physical Systems, Concepts in Solid State Physics.

Experimental Physics: General Physics Laboratory, Intermediate Mechanics, Oscillations and Waves Laboratory, Intermediate Electromagnetism Laboratory, Intermediate Thermal Physics Laboratory.

Quantum Technology Minor: Introduction to Quantum Computation, Introduction to Quantum Communications and Cryptography, Introduction to Quantum Measurement, Introduction to Materials for Quantum Technologies, Quantum Optics.

Honours And Awards

QC RISE Internship Program (Sep 2026 – Feb 2027) Year: 2026

- Selected for QC RISE (under National Quantum Mission) fellowship in rapidly advancing field of Quantum Computing.

S.N Bhatt Memorial Excellence Fellowship 2026 Year: 2026

- Selected for the ICTS–S. N. Bhatt Memorial Excellence Fellowship (2026), a competitive research program at the International Centre for Theoretical Sciences (TIFR).

Mallika science fellowship for four years (2022-26) of B.S. (Research). Year: 2022

- For female students with good rank in highly competitive exams of the country.

International Astronomy and Astrophysics Competition (IAAC)

- Achieved top 7% ranking internationally in the final round of the IAAC awarded Silver Honour. Year: 2024

International Physicists Tournament (IPT) Year: 2024

- Obtained Rank 3 in India at the International Physicists Tournament (IPT) national selections.
- Wrote Paper along with 5 teammates on "Varying parameters (e.g., peg size, distribution, bead format in Galton Board) and introducing vibrations can yield distributions that deviate from binomial and may not adhere to the central limit theorem in an independent and identical distributed scenario".
- Learnt 3D modelling and 3D printing.
- Represented IISc at national level.

IIT JEE ADVANCED - 2022 All India Rank 3772 Year: 2022

Indian Institute of Technology – Joint Entrance Examination (IIT-JEE)
Joint Admission Board (JAB), Government of India

- Applicants: Approximately 155,538 students (JEE MAIN Qualified students out of 1,026,799 students).
- IIT-JEE of India is listed as the second in the toughest exam in the world's list.

JEE MAIN Percentile Rank - 98.87% Year: 2022

National Testing Agency (NTA), Government of India

- Secured 98.875 percentile Rank.
- Applicants: Approximately 1,026,799 students.

LIC Student of the Year Award in 11th standard Year: 2021

Workshops and Conferences

- Participated and Volunteered for TQC 2025 Conference (Theory of Quantum Computation, Communication and Cryptography (TQC)) held at IISc, Bengaluru.
- Presented a poster in ASI 2025 (Astronomical Society of India) conference.
  Abstract "Systematics of the Multi-Frequency Properties of Accreting Supermassive Black Holes in the Local Universe"
  https://ui.adsabs.harvard.edu/abs/2025asi..confP..97P/abstract
- Selected and attended "Summer School in Optics and Photonics'24" at IISc.
- Selected and Attended for Summer School for Women in Physics, 2024 at ICTS (International Centre for Theoretical Sciences, Bengaluru) from 3rd June to 14th June 2024.
  (one of the 65 students to be selected in the entire country (India))
- "Beginning Astronomy v3: Start a data-driven journey" workshop held on 11–13 Dec 2023 at IIT Hyderabad by IUCAA.
  - One of the 35 students selected in the entire country (India).
  - Learnt to access telescopes, satellite, and Solar Dynamics Observatory data and to analyse it using Python and SAO Image DS9 software.
  - Learnt about solar physics and formation of sunspots, Pulsar, Neutron stars, and LIGO detection of gravitational waves.

Certificates /Online Courses

- "From Big Bang to Dark Energy" by the University of Tokyo (organised by Hitoshi Murayama).
- Certificate course by MATLAB Onramp.
- "Elements of AI" by University of Helsinki (2 ECTS credits).
- "How Things work: An Introduction to physics" by the University of Virginia.
- Qualified the SCRATCH assessment by Google, conducted as a part of Google India Code (2018 training program).
- Certificate by ISRO for participating in an exhibition and explaining about rockets and satellite to the public.

Student Membership

Convenor of QUARKS Magazine (Ug club) – Core Member Year: 2026

Convenor of ASTRAE CLUB (Indian Institute of Science Astronomy club) Year: 2024
(Organised Talks, Night Sky observation Sessions using Telescopes, Astrophotography)

Technical Skills

- Efficient in Python, SQL, MATLAB, Mathematica, LaTeX.
- Comfortable with QISKIT and QuTip.
- Efficient in telescope handling (Dobsonium, Cassegrain, Maksutov).
- Familiar with the basics of designing 3D models (in Fusion 360 software) and 3D printing.
- Learned astrophotography and image processing using Siril Software.

Teaching & Academic Outreach

Physics and Mathematics Tutor, QUAZAR February 2026 – Present

quazaredu.com

- Tutor students in physics and mathematics, with a focus on building conceptual understanding and problem-solving skills.
- Mentor students preparing for competitive examinations and olympiads, including NSEP and Physics Olympiad-level preparation.
- Occasionally volunteer on Saturdays, when time permits, to teach English to children in government schools as part of IISc's Notebook Drive outreach initiative.

Co-curricular Activities

- **Reading:** I Enjoy literary fiction, contemporary novels, and psychological thrillers, including works such as A Thousand Splendid Suns, A Little Life, and The Silent Patient. (goodreads)
- **Trekking and Travel:** Passionate about travelling and trekking; completed several treks, including moderate-to-difficult Himalayan routes such as the Kartik Swami Temple trek and the Dhanoti 2/Kug Pass route in Himachal Pradesh (India).
- **Snooker:** Senior snooker player; won multiple medals in IISc tournaments, including undergraduate and overall-category events.
- **Writing:** Occasionally write blogs and poems; core member of QUARKS, the IISc undergraduate magazine club.
- **Painting and Sketching:** Practising canvas painting and sketching since childhood; won state-level and other art competitions.

References

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